

term0

$$+ \left(\frac{1}{24} * a_t^2 - \frac{1}{4} * a_t * \hat{\mu} + \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} - \frac{1}{2} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon^{\alpha_3 \alpha_4} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \cdot (H)_{\alpha_1 \alpha_2, i}^{\dot{\alpha}_1 \dot{\alpha}_2} (H^\dagger)_{\alpha_3 \alpha_4, j}^{\dot{\alpha}_3 \dot{\alpha}_4} \quad (1)$$

term1

$$+ \left(a_t * \hat{\mu} - \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{2} * a_t^2 + \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + \frac{5}{2} * a_t^2 * \hat{\mu}_h^2 + 2 * \frac{a_t * \hat{\mu}}{\varepsilon} + 2 * \frac{a_t^2 * \hat{\mu}_h^2}{\varepsilon} \right) * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_2} \quad (2)$$

term2

$$+ \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{2} * a_t * \hat{\mu} + \frac{a_t * \hat{\mu}}{\varepsilon} \right) * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_{\alpha_1 \alpha_2, i}^{\dot{\alpha}_1 \dot{\alpha}_2} (H^\dagger)_j \quad (3)$$

term3

$$- \left(\frac{1}{4} * a_t * \hat{\mu} - \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{2} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon^{\alpha_3 \alpha_4} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \cdot (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2 \alpha_3 \alpha_4, j}^{\dot{\alpha}_2 \dot{\alpha}_3 \dot{\alpha}_4} \quad (4)$$

term4

$$- \left(\frac{1}{4} * a_t * \hat{\mu} - \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{2} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon^{\alpha_3 \alpha_4} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \cdot (H)_{\alpha_1 \alpha_2 \alpha_3, i}^{\dot{\alpha}_1 \dot{\alpha}_2 \dot{\alpha}_3} (H^\dagger)_{\alpha_4, j}^{\dot{\alpha}_4} \quad (5)$$

term5

$$- \left(\frac{1}{2} * a_t * \hat{\mu} - \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon^{\alpha_3 \alpha_4} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \cdot (H)_{\alpha_1 \alpha_2, i}^{\dot{\alpha}_1 \dot{\alpha}_2} (H^\dagger)_{\alpha_3 \alpha_4, j}^{\dot{\alpha}_3 \dot{\alpha}_4} \quad (6)$$

term6

$$- \left(\frac{1}{8} * a_t * \hat{\mu} - \frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{4} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon^{\alpha_3 \alpha_4} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \cdot (H)_{\alpha_1 \alpha_2 \alpha_3 \alpha_4, i}^{\dot{\alpha}_1 \dot{\alpha}_2 \dot{\alpha}_3 \dot{\alpha}_4} (H^\dagger)_j \quad (7)$$

term7

$$- \left(\frac{1}{4} * a_t * \hat{\mu} - \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{2} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon^{\alpha_3 \alpha_4} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \cdot (H)_{\alpha_1 \alpha_2 \alpha_3, i}^{\dot{\alpha}_1 \dot{\alpha}_2 \dot{\alpha}_3} (H^\dagger)_{\alpha_4, j}^{\dot{\alpha}_4} \quad (8)$$

term8

$$\begin{aligned}
& - \left(\frac{1}{4} * a_t * \hat{\mu} - \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{2} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} \\
& * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon^{\alpha_3 \alpha_4} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \cdot (H)_{\alpha_1, i} (H^\dagger)_{\alpha_2 \alpha_3 \alpha_4, j}^{\dot{\alpha}_1 \dot{\alpha}_2 \dot{\alpha}_3 \dot{\alpha}_4}
\end{aligned} \tag{9}$$

term9

$$+ \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{2} * a_t * \hat{\mu} + \frac{a_t * \hat{\mu}}{\varepsilon} \right) * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_i (H^\dagger)_{\alpha_1 \alpha_2, j}^{\dot{\alpha}_1 \dot{\alpha}_2} \tag{10}$$

term10

$$\begin{aligned}
& - \left(\frac{1}{8} * a_t * \hat{\mu} - \frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{4} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} \\
& * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon^{\alpha_3 \alpha_4} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \cdot (H)_i (H^\dagger)_{\alpha_1 \alpha_2 \alpha_3 \alpha_4, j}^{\dot{\alpha}_1 \dot{\alpha}_2 \dot{\alpha}_3 \dot{\alpha}_4}
\end{aligned} \tag{11}$$

term11

$$\begin{aligned}
& + \left(a_t^2 + a_t^2 * \hat{\mu}_h^2 + a_t^2 * \hat{\mu}_h^4 - \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^4 - \frac{1}{2} * a_t * \hat{\mu} + \frac{1}{2} \right. \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{2} * \kappa - \frac{1}{2} * \kappa * \ln \frac{M_s^2}{\mu_M^2} + 2 * \frac{a_t^2}{\varepsilon} + 2 * \frac{a_t^2 * \hat{\mu}_h^2}{\varepsilon} + 2 * \frac{a_t^2 * \hat{\mu}_h^4}{\varepsilon} - \frac{a_t * \hat{\mu}}{\varepsilon} + \frac{\kappa}{\varepsilon} \Big) \\
& * M_s^2 * \varepsilon_{ij} \cdot (H)_i (H^\dagger)_j
\end{aligned} \tag{12}$$

term12

$$\begin{aligned}
& + \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \cdot (H)_{\alpha_1, i} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_1 \dot{\alpha}_2 \dot{\alpha}_3 \dot{\alpha}_4} (B_R)^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& + \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_4} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_2} \cdot (H)_{\alpha_1, i} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_1 \dot{\alpha}_2 \dot{\alpha}_3 \dot{\alpha}_4} (B_R)^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& - \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_3 \alpha_1} \varepsilon^{\alpha_4 \alpha_2} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_4} \cdot (B_L)_{\alpha_3 \alpha_4} (H)_{\alpha_1, i} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_1 \dot{\alpha}_2 \dot{\alpha}_3 \dot{\alpha}_4} \\
& - \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_3 \alpha_2} \varepsilon^{\alpha_4 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_4} \cdot (B_L)_{\alpha_3 \alpha_4} (H)_{\alpha_1, i} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_1 \dot{\alpha}_2 \dot{\alpha}_3 \dot{\alpha}_4}
\end{aligned} \tag{13}$$

term13

$$\begin{aligned}
& + \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \cdot (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_2} (B_R)^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& + \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_4} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_2} \cdot (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_2} (B_R)^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& - \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_3 \alpha_1} \varepsilon_{\dot{\alpha}_4 \alpha_2} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_4} \cdot (B_L)_{\alpha_3 \alpha_4} (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_2} \\
& - \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_3 \alpha_2} \varepsilon_{\dot{\alpha}_4 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_4} \cdot (B_L)_{\alpha_3 \alpha_4} (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_2}
\end{aligned} \tag{14}$$

term14

$$\begin{aligned}
& + \left(a_t^3 * \hat{\mu} * \hat{\mu}_h^2 - a_t^4 * \hat{\mu}_h^2 + \ln \frac{M_s^2}{\mu_M^2} * a_t^4 * \hat{\mu}_h^2 - \kappa * a_t^2 - \kappa * a_t^2 * \hat{\mu}_h^2 - \kappa * a_t^2 * \hat{\mu}_h^4 + \kappa \right. \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^4 + \frac{3}{32} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{8} \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 + \frac{1}{8} * \lambda_\varphi * a_t^2 - \frac{1}{8} * \lambda_\varphi * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{9}{64} * g_2^2 * a_t^2 - \frac{3}{16} * \frac{g_2^2 * a_t^2}{\varepsilon} \\
& - \frac{1}{4} * a_t^2 * \hat{\mu}^2 - \frac{1}{4} * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} - \frac{1}{4} * \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} + \frac{1}{4} * \frac{\lambda_\varphi * a_t^2}{\varepsilon} + \frac{1}{2} * a_t^3 * \hat{\mu} - \frac{1}{2} \\
& * a_t^4 - \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 + \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 + \frac{1}{2} * \kappa * a_t * \hat{\mu} + \frac{1}{2} * \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} \\
& \left. - 2 * \frac{a_t^4 * \hat{\mu}_h^2}{\varepsilon} - 2 * \frac{\kappa * a_t^2}{\varepsilon} - 2 * \frac{\kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - 2 * \frac{\kappa * a_t^2 * \hat{\mu}_h^4}{\varepsilon} + \frac{a_t^3 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} - \frac{a_t^4}{\varepsilon} \right) \\
& * M_s^{-2} * \varepsilon_{jk} \varepsilon_{il} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_{\alpha_1 \alpha_2, i}^{\dot{\alpha}_1 \dot{\alpha}_2} (H)_j (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{15}$$

term15

$$\begin{aligned}
& - \left(\frac{3}{32} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 - \frac{1}{8} * \lambda_\varphi * a_t^2 + \frac{1}{8} * \lambda_\varphi * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{9}{64} \right. \\
& * g_2^2 * a_t^2 - \frac{3}{16} * \frac{g_2^2 * a_t^2}{\varepsilon} + \frac{1}{4} * a_t^2 * \hat{\mu}^2 - \frac{1}{4} * \kappa * a_t * \hat{\mu} + \frac{1}{4} * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{4} * \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} \\
& \left. - \frac{1}{4} * \frac{\lambda_\varphi * a_t^2}{\varepsilon} - \frac{1}{2} * \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} * \varepsilon_{jl} \varepsilon_{ik} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_{\alpha_1 \alpha_2, i}^{\dot{\alpha}_1 \dot{\alpha}_2} (H)_j (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{16}$$

term16

$$\begin{aligned}
& + \left(a_t^3 * \hat{\mu} * \hat{\mu}_h^2 - a_t^4 * \hat{\mu}_h^2 + \ln \frac{M_s^2}{\mu_M^2} * a_t^4 * \hat{\mu}_h^2 - \kappa * a_t^2 - \kappa * a_t^2 * \hat{\mu}_h^2 - \kappa * a_t^2 * \hat{\mu}_h^4 + \kappa \right. \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^4 + \frac{3}{32} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{8} \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 + \frac{1}{8} * \lambda_\varphi * a_t^2 - \frac{1}{8} * \lambda_\varphi * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{9}{64} * g_2^2 * a_t^2 - \frac{3}{16} * \frac{g_2^2 * a_t^2}{\varepsilon} \\
& - \frac{1}{4} * a_t^2 * \hat{\mu}^2 - \frac{1}{4} * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} - \frac{1}{4} * \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} + \frac{1}{4} * \frac{\lambda_\varphi * a_t^2}{\varepsilon} + \frac{1}{2} * a_t^3 * \hat{\mu} - \frac{1}{2} \\
& * a_t^4 - \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 + \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 + \frac{1}{2} * \kappa * a_t * \hat{\mu} + \frac{1}{2} * \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} \\
& - 2 * \frac{a_t^4 * \hat{\mu}_h^2}{\varepsilon} - 2 * \frac{\kappa * a_t^2}{\varepsilon} - 2 * \frac{\kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - 2 * \frac{\kappa * a_t^2 * \hat{\mu}_h^4}{\varepsilon} + \frac{a_t^3 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} - \frac{a_t^4}{\varepsilon} \Big) \\
& * M_s^{-2} * \varepsilon_{jk} \varepsilon_{il} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_i (H)_j (H^\dagger)_{\alpha_1 \alpha_2, k}^{\dot{\alpha}_1 \dot{\alpha}_2} (H^\dagger)_l
\end{aligned} \tag{17}$$

term17

$$\begin{aligned}
& - \left(\frac{3}{32} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 - \frac{1}{8} * \lambda_\varphi * a_t^2 + \frac{1}{8} * \lambda_\varphi * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{9}{64} \right. \\
& * g_2^2 * a_t^2 - \frac{3}{16} * \frac{g_2^2 * a_t^2}{\varepsilon} + \frac{1}{4} * a_t^2 * \hat{\mu}^2 - \frac{1}{4} * \kappa * a_t * \hat{\mu} + \frac{1}{4} * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{4} * \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} \\
& \left. - \frac{1}{4} * \frac{\lambda_\varphi * a_t^2}{\varepsilon} - \frac{1}{2} * \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} * \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_i (H)_j (H^\dagger)_{\alpha_1 \alpha_2, k}^{\dot{\alpha}_1 \dot{\alpha}_2} (H^\dagger)_l
\end{aligned} \tag{18}$$

term18

$$\begin{aligned}
& - \left(a_t^2 * \hat{\mu}^2 + \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} - \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 \right. \\
& + \frac{1}{24} * d * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{11}{144} * d * g_2^2 * a_t^2 - \frac{1}{12} * \frac{d * g_2^2 * a_t^2}{\varepsilon} - \frac{1}{8} * \varepsilon * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^3 * \hat{\mu} + \frac{1}{8} * \varepsilon * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{4} * \varepsilon * a_t^3 * \hat{\mu} - \frac{1}{4} * \varepsilon * \kappa * a_t^2 - \frac{29}{72} * g_2^2 * a_t^2 \\
& - \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 - \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 - \frac{1}{2} * \kappa * a_t^2 * \hat{\mu}_h^2 - \frac{1}{2} * \lambda_\varphi * a_t^2 + \frac{1}{2} * \lambda_\varphi \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{2} * g_2^2 * a_t^2 * \hat{\mu}_h^2 + \frac{1}{2} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{1}{2} * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 \\
& + \frac{1}{2} * \varepsilon * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 + \frac{7}{12} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{4} * a_t^3 * \hat{\mu} - \frac{3}{4} * \varepsilon * a_t^4 - \frac{7}{6} * \frac{g_2^2 * a_t^2}{\varepsilon} \\
& - \frac{5}{4} * a_t^4 + \frac{5}{4} * \kappa * a_t^2 - \frac{3}{2} * \kappa * a_t * \hat{\mu} + \frac{9}{4} * \lambda_h * a_t^2 - 2 * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 + 2 * a_t^4 * \hat{\mu}_h^2 - 2 \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 * \hat{\mu}_h^2 + 2 * \kappa * a_t^2 * \hat{\mu}_h^4 - 2 * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - 2 * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^4 \\
& - 2 * \frac{a_t^3 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} - 2 * \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} + 2 * \frac{\kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + 4 * \frac{a_t^4 * \hat{\mu}_h^2}{\varepsilon} + 4 * \frac{\kappa * a_t^2}{\varepsilon} \\
& + 4 * \frac{\kappa * a_t^2 * \hat{\mu}_h^4}{\varepsilon} + \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} + \frac{a_t^4}{\varepsilon} - \frac{\lambda_\varphi * a_t^2}{\varepsilon} - \frac{g_2^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + \frac{\lambda_h * a_t^2}{\varepsilon} \Big) \\
& * M_s^{-2} * \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H)_j (H^\dagger)_{\alpha_2, k}^{\dot{\alpha}_2} (H^\dagger)_l
\end{aligned} \tag{19}$$

term19

$$\begin{aligned}
& + \left(\frac{1}{96} * d * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{11}{576} * d * g_1^2 * a_t^2 - \frac{1}{48} * \kappa^2 + \frac{1}{48} * d * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 \right. \\
& - \frac{1}{48} * \frac{d * g_1^2 * a_t^2}{\varepsilon} - \frac{11}{288} * d * g_2^2 * a_t^2 - \frac{1}{24} * g_1^2 * a_t^2 - \frac{1}{24} * \frac{d * g_2^2 * a_t^2}{\varepsilon} - \frac{5}{96} * g_2^2 \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{5}{48} * \frac{g_2^2 * a_t^2}{\varepsilon} + \frac{1}{8} * g_2^2 * a_t^2 * \hat{\mu}_h^2 - \frac{1}{8} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{1}{8} * g_1^2 * a_t^2 \\
& * \hat{\mu}_h^2 + \frac{1}{8} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{8} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{1}{6} * a_t^3 * \hat{\mu} - \frac{3}{16} * \varepsilon * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^4 + \frac{121}{576} * g_2^2 * a_t^2 + \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 - \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 + \frac{1}{4} * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{4} \\
& * \frac{g_2^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - \frac{1}{4} * \frac{g_1^2 * a_t^2}{\varepsilon} - \frac{1}{4} * \frac{g_1^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - \frac{13}{48} * a_t^2 * \hat{\mu}^2 + \frac{11}{32} * \varepsilon * a_t^4 + \frac{5}{12} \\
& * \kappa * a_t^2 - \frac{1}{2} * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} - \frac{1}{2} * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{1}{2} * \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} + \frac{1}{2} * \frac{a_t^4}{\varepsilon} \\
& - \frac{1}{2} * \frac{\lambda_h * a_t^2}{\varepsilon} + \frac{13}{24} * \kappa * a_t * \hat{\mu} - \frac{7}{8} * \lambda_h * a_t^2 + \frac{7}{6} * a_t^4 + \frac{5}{4} * \kappa * a_t^2 * \hat{\mu}_h^2 + \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} \\
& + \frac{\kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \Big) * M_s^{-2} * \varepsilon_{jl} \varepsilon_{ik} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H)_{\alpha_2, j}^{\dot{\alpha}_2} (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{20}$$

term20

$$\begin{aligned}
& - \left(\kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{1}{48} * d * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{48} * d * g_1^2 \right. \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{11}{288} * d * g_2^2 * a_t^2 - \frac{11}{288} * d * g_1^2 * a_t^2 + \frac{1}{24} * \kappa^2 + \frac{1}{24} * \frac{d * g_2^2 * a_t^2}{\varepsilon} - \frac{1}{24} \\
& * \frac{d * g_1^2 * a_t^2}{\varepsilon} + \frac{29}{144} * g_2^2 * a_t^2 - \frac{29}{144} * g_1^2 * a_t^2 + \frac{1}{4} * g_2^2 * a_t^2 * \hat{\mu}_h^2 - \frac{1}{4} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 \\
& * \hat{\mu}_h^2 - \frac{1}{4} * g_1^2 * a_t^2 * \hat{\mu}_h^2 + \frac{1}{4} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{7}{24} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{7}{24} * g_1^2 \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{3} * a_t^3 * \hat{\mu} + \frac{3}{8} * \varepsilon * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 - \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 + \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 \\
& - \frac{1}{2} * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{2} * \frac{g_2^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - \frac{1}{2} * \frac{g_1^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + \frac{13}{24} * a_t^2 * \hat{\mu}^2 + \frac{7}{12} \\
& * \frac{g_2^2 * a_t^2}{\varepsilon} - \frac{7}{12} * \frac{g_1^2 * a_t^2}{\varepsilon} - \frac{11}{16} * \varepsilon * a_t^4 - \frac{5}{6} * \kappa * a_t^2 - \frac{13}{12} * \kappa * a_t * \hat{\mu} + \frac{9}{4} * \lambda_h * a_t^2 - \frac{7}{3} \\
& * a_t^4 - \frac{5}{2} * \kappa * a_t^2 * \hat{\mu}_h^2 - 2 * \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} - 2 * \frac{\kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} - \frac{a_t^4}{\varepsilon} + \frac{\lambda_h * a_t^2}{\varepsilon} \Big) \\
& * M_s^{-2} * \varepsilon_{il} \varepsilon_{jk} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H)_j (H^\dagger)_{\alpha_2, k}^{\dot{\alpha}_2} (H^\dagger)_l
\end{aligned} \tag{21}$$

term21

$$\begin{aligned}
& - \left(\frac{1}{96} * d * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{11}{576} * d * g_2^2 * a_t^2 - \frac{1}{48} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{d * g_2^2 * a_t^2}{\varepsilon} \right. \\
& + \frac{1}{24} * \frac{g_1^2 * a_t^2}{\varepsilon} + \frac{17}{288} * g_1^2 * a_t^2 - \frac{19}{96} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{4} * g_2^2 * a_t^2 * \hat{\mu}_h^2 - \frac{1}{4} * g_2^2 \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + \frac{1}{4} * \lambda_h * a_t^2 + \frac{179}{576} * g_2^2 * a_t^2 + \frac{19}{48} * \frac{g_2^2 * a_t^2}{\varepsilon} + \frac{1}{2} * \frac{g_2^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \Big) \\
& * M_s^{-2} * \varepsilon_{jk} \varepsilon_{il} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H)_{\alpha_2, j}^{\dot{\alpha}_2} (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{22}$$

term22

$$\begin{aligned}
& + \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \delta_{ik} \delta_{lj} \cdot (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_2} (W_R)_{kl}^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& + \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \delta_{ik} \delta_{lj} \cdot (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_2} (W_R)_{kl}^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& - \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon^{\alpha_3 \alpha_1} \varepsilon^{\alpha_4 \alpha_2} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_4} \delta_{ik} \delta_{lj} \cdot (W_L)_{\alpha_3 \alpha_4, kl} (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_2} \\
& - \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon^{\alpha_3 \alpha_2} \varepsilon^{\alpha_4 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_4} \delta_{ik} \delta_{lj} \cdot (W_L)_{\alpha_3 \alpha_4, kl} (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_2}
\end{aligned} \tag{23}$$

term23

$$\begin{aligned}
& + \left(\frac{1}{96} * d * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{11}{576} * d * g_1^2 * a_t^2 - \frac{1}{48} * \kappa^2 + \frac{1}{48} * d * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 \right. \\
& - \frac{1}{48} * \frac{d * g_1^2 * a_t^2}{\varepsilon} - \frac{11}{288} * d * g_2^2 * a_t^2 - \frac{1}{24} * g_1^2 * a_t^2 - \frac{1}{24} * \frac{d * g_2^2 * a_t^2}{\varepsilon} - \frac{5}{96} * g_2^2 \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{5}{48} * \frac{g_2^2 * a_t^2}{\varepsilon} + \frac{1}{8} * g_2^2 * a_t^2 * \hat{\mu}_h^2 - \frac{1}{8} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{1}{8} * g_1^2 * a_t^2 \\
& * \hat{\mu}_h^2 + \frac{1}{8} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{8} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{1}{6} * a_t^3 * \hat{\mu} - \frac{3}{16} * \varepsilon * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^4 + \frac{121}{576} * g_2^2 * a_t^2 + \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 - \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 + \frac{1}{4} * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{4} \\
& * \frac{g_2^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - \frac{1}{4} * \frac{g_1^2 * a_t^2}{\varepsilon} - \frac{1}{4} * \frac{g_1^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - \frac{13}{48} * a_t^2 * \hat{\mu}^2 + \frac{11}{32} * \varepsilon * a_t^4 + \frac{5}{12} \\
& * \kappa * a_t^2 - \frac{1}{2} * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} - \frac{1}{2} * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{1}{2} * \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} + \frac{1}{2} * \frac{a_t^4}{\varepsilon} \\
& - \frac{1}{2} * \frac{\lambda_h * a_t^2}{\varepsilon} + \frac{13}{24} * \kappa * a_t * \hat{\mu} - \frac{7}{8} * \lambda_h * a_t^2 + \frac{7}{6} * a_t^4 + \frac{5}{4} * \kappa * a_t^2 * \hat{\mu}_h^2 + \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} \\
& \left. + \frac{\kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \right) * M_s^{-2} * \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_i (H)_j (H^\dagger)_{\alpha_1, k}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2, l}^{\dot{\alpha}_2}
\end{aligned} \tag{24}$$

term24

$$\begin{aligned}
& - \left(\frac{1}{96} * d * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{11}{576} * d * g_2^2 * a_t^2 - \frac{1}{48} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{d * g_2^2 * a_t^2}{\varepsilon} \right. \\
& + \frac{1}{24} * \frac{g_1^2 * a_t^2}{\varepsilon} + \frac{17}{288} * g_1^2 * a_t^2 - \frac{19}{96} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{4} * g_2^2 * a_t^2 * \hat{\mu}_h^2 - \frac{1}{4} * g_2^2 \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + \frac{1}{4} * \lambda_h * a_t^2 + \frac{179}{576} * g_2^2 * a_t^2 + \frac{19}{48} * \frac{g_2^2 * a_t^2}{\varepsilon} + \frac{1}{2} * \frac{g_2^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \Big) \\
& * M_s^{-2} * \varepsilon_{jk} \varepsilon_{il} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_i (H)_j (H^\dagger)_{\alpha_1, k}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2, l}^{\dot{\alpha}_2}
\end{aligned} \tag{25}$$

term25

$$\begin{aligned}
& + \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \delta_{ik} \delta_{lj} \cdot (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_2} (W_R)_{kl}^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& + \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_4} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_2} \delta_{ik} \delta_{lj} \cdot (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_2} (W_R)_{kl}^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& - \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon^{\alpha_3 \alpha_1} \varepsilon^{\alpha_4 \alpha_2} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_4} \delta_{ik} \delta_{lj} \cdot (W_L)_{\alpha_3 \alpha_4, kl} (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_2} \\
& - \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) \\
& * M_s^{-2} * \varepsilon^{\alpha_3 \alpha_2} \varepsilon^{\alpha_4 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_4} \delta_{ik} \delta_{lj} \cdot (W_L)_{\alpha_3 \alpha_4, kl} (H)_{\alpha_1, i}^{\dot{\alpha}_1} (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_2}
\end{aligned} \tag{26}$$

term26

$$\begin{aligned}
& + \left(a_t^3 * \hat{\mu} - \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 - \frac{1}{4} * a_t^2 * \hat{\mu}^2 - \frac{1}{4} * \kappa^2 * \ln \frac{M_s^2}{\mu_M^2} + \frac{1}{4} * \lambda_\varphi * a_t^2 - \frac{1}{4} \right. \\
& * \lambda_\varphi * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{2} * \kappa * a_t * \hat{\mu} + \frac{1}{2} * \frac{\kappa^2}{\varepsilon} + \frac{1}{2} * \frac{\lambda_\varphi * a_t^2}{\varepsilon} + 2 * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 + 2 \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 - 2 * \kappa * a_t^2 * \hat{\mu}_h^4 + 2 * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + 2 * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^4 + 2 \\
& * \frac{a_t^3 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} - 3 * a_t^4 - 3 * \kappa * a_t^2 + 3 * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + 3 * \lambda_h * a_t^2 - 3 * \lambda_h \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - 4 * \kappa * a_t^2 * \hat{\mu}_h^2 - 4 * \frac{a_t^4}{\varepsilon} - 4 * \frac{\kappa * a_t^2}{\varepsilon} - 4 * \frac{\kappa * a_t^2 * \hat{\mu}_h^4}{\varepsilon} + 6 * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^4 * \hat{\mu}_h^2 + 6 * \lambda_h * a_t^2 * \hat{\mu}_h^2 - 6 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - 6 * \frac{\kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + 6 * \frac{\lambda_h * a_t^2}{\varepsilon} \\
& \left. - 8 * a_t^4 * \hat{\mu}_h^2 - 12 * \frac{a_t^4 * \hat{\mu}_h^2}{\varepsilon} + 12 * \frac{\lambda_h * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \right) * \varepsilon_{ik} \varepsilon_{jl} \cdot (H)_i (H)_j (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{27}$$

term27

$$\begin{aligned}
& - \left(\frac{5}{576} * g_1^2 * a_t^2 - \frac{1}{96} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{48} * \frac{g_1^2 * a_t^2}{\varepsilon} \right) \\
& * M_s^{-2} * \varepsilon_{ij} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \cdot (H)_i (H^\dagger)_j (B_R)^{\dot{\alpha}_1 \dot{\alpha}_2} (B_R)^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& - \left(\frac{5}{576} * g_1^2 * a_t^2 - \frac{1}{96} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{48} * \frac{g_1^2 * a_t^2}{\varepsilon} \right) \\
& * M_s^{-2} * \varepsilon_{ij} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_4} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_2} \cdot (H)_i (H^\dagger)_j (B_R)^{\dot{\alpha}_1 \dot{\alpha}_2} (B_R)^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& - \left(\frac{5}{576} * g_1^2 * a_t^2 - \frac{1}{96} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{48} * \frac{g_1^2 * a_t^2}{\varepsilon} \right) \\
& * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon^{\alpha_3 \alpha_4} \cdot (B_L)_{\alpha_1 \alpha_2} (B_L)_{\alpha_3 \alpha_4} (H)_i (H^\dagger)_j \\
& - \left(\frac{5}{576} * g_1^2 * a_t^2 - \frac{1}{96} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{48} * \frac{g_1^2 * a_t^2}{\varepsilon} \right) \\
& * M_s^{-2} * \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_4} \varepsilon^{\alpha_3 \alpha_2} \cdot (B_L)_{\alpha_1 \alpha_2} (B_L)_{\alpha_3 \alpha_4} (H)_i (H^\dagger)_j
\end{aligned} \tag{28}$$

term28

$$\begin{aligned}
& + \left(\frac{5}{288} * g_1 * g_2 * a_t^2 - \frac{1}{48} * g_1 * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{24} * \frac{g_1 * g_2 * a_t^2}{\varepsilon} \right) \\
& * M_s^{-2} * \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \delta_{ik} \delta_{lj} \cdot (H)_i (H^\dagger)_j (B_R)^{\dot{\alpha}_1 \dot{\alpha}_2} (W_R)_{kl}^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& + \left(\frac{5}{288} * g_1 * g_2 * a_t^2 - \frac{1}{48} * g_1 * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{24} * \frac{g_1 * g_2 * a_t^2}{\varepsilon} \right) \\
& * M_s^{-2} * \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_4} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_2} \delta_{ik} \delta_{lj} \cdot (H)_i (H^\dagger)_j (B_R)^{\dot{\alpha}_1 \dot{\alpha}_2} (W_R)_{kl}^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& + \left(\frac{5}{288} * g_1 * g_2 * a_t^2 - \frac{1}{48} * g_1 * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{24} * \frac{g_1 * g_2 * a_t^2}{\varepsilon} \right) \\
& * M_s^{-2} * \varepsilon^{\alpha_1 \alpha_2} \varepsilon^{\alpha_3 \alpha_4} \delta_{ik} \delta_{lj} \cdot (B_L)_{\alpha_1 \alpha_2} (W_L)_{\alpha_3 \alpha_4, kl} (H)_i (H^\dagger)_j \\
& + \left(\frac{5}{288} * g_1 * g_2 * a_t^2 - \frac{1}{48} * g_1 * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{24} * \frac{g_1 * g_2 * a_t^2}{\varepsilon} \right) \\
& * M_s^{-2} * \varepsilon^{\alpha_1 \alpha_4} \varepsilon^{\alpha_3 \alpha_2} \delta_{ik} \delta_{lj} \cdot (B_L)_{\alpha_1 \alpha_2} (W_L)_{\alpha_3 \alpha_4, kl} (H)_i (H^\dagger)_j
\end{aligned} \tag{29}$$

term29

$$\begin{aligned}
& + \left(a_t^4 * \hat{\mu}^2 - \kappa * a_t^3 * \hat{\mu} * \hat{\mu}_h^4 + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 \right. \\
& * \hat{\mu} * \hat{\mu}_h^4 - \lambda_\varphi * a_t^4 * \hat{\mu}_h^2 - \frac{1}{12} * \kappa^3 - \frac{1}{12} * \lambda_\varphi * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} - \frac{1}{6} * a_t^3 \\
& * \hat{\mu}^3 + \frac{1}{6} * \frac{\lambda_\varphi * a_t^3 * \hat{\mu}}{\varepsilon} - \frac{1}{4} * \kappa^2 * a_t * \hat{\mu} + \frac{1}{4} * \lambda_\varphi * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 \\
& + \frac{1}{3} * \lambda_\varphi * a_t^3 * \hat{\mu} + \frac{1}{2} * \kappa * a_t^2 * \hat{\mu}^2 - \frac{1}{2} * \lambda_\varphi * a_t^4 + \frac{1}{2} * \lambda_\varphi * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^4 * \hat{\mu}_h^2 - \frac{1}{2} * \lambda_\varphi * \kappa * a_t^2 - \frac{1}{2} * \frac{\lambda_\varphi * \kappa * a_t^2}{\varepsilon} - \frac{3}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 * \hat{\mu}^2 \\
& * \hat{\mu}_h^2 + \frac{7}{2} * a_t^4 * \hat{\mu}^2 * \hat{\mu}_h^2 + \frac{7}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^5 * \hat{\mu} + \frac{11}{2} * \kappa^2 * a_t^2 - \frac{15}{2} * a_t^5 \\
& * \hat{\mu} + \frac{19}{2} * \kappa^2 * a_t^2 * \hat{\mu}_h^2 - \frac{21}{2} * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 + \frac{43}{3} * a_t^6 + \frac{37}{2} * \kappa * a_t^4 \\
& - 2 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} - 2 * \frac{\kappa * a_t^3 * \hat{\mu}}{\varepsilon} - 2 * \frac{\kappa * a_t^3 * \hat{\mu} * \hat{\mu}_h^4}{\varepsilon} + 3 \\
& * a_t^6 * \hat{\mu}_h^2 - 3 * \ln \frac{M_s^2}{\mu_M^2} * a_t^6 * \hat{\mu}_h^2 + 3 * \kappa^2 * a_t^2 * \hat{\mu}_h^4 - 3 * \kappa^2 * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^2 - 3 * \kappa^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^4 + 3 * \frac{a_t^4 * \hat{\mu}^2 * \hat{\mu}_h^2}{\varepsilon} + 4 * \lambda_h * a_t^3 * \hat{\mu} \\
& + 4 * \lambda_h^2 * a_t^2 - 4 * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + 4 * \frac{\lambda_h * a_t^3 * \hat{\mu}}{\varepsilon} + 5 * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^5 * \hat{\mu} * \hat{\mu}_h^2 - 5 * \kappa * a_t^3 * \hat{\mu} + 6 * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 - 6 * \kappa^2 \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + 6 * \frac{a_t^6 * \hat{\mu}_h^2}{\varepsilon} + 6 * \frac{\kappa^2 * a_t^2}{\varepsilon} + 6 * \frac{\kappa^2 * a_t^2 * \hat{\mu}_h^4}{\varepsilon} \\
& - 7 * a_t^5 * \hat{\mu} * \hat{\mu}_h^2 - 7 * \frac{a_t^5 * \hat{\mu}}{\varepsilon} - 8 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 + 8 * \lambda_h \\
& * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + 8 * \frac{\lambda_h^2 * a_t^2}{\varepsilon} - 9 * \ln \frac{M_s^2}{\mu_M^2} * a_t^6 - 10 * \lambda_h * \kappa * a_t^2 \\
& - 10 * \frac{a_t^5 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} - 12 * \kappa * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 + 12 * \lambda_h * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 - 12 \\
& * \lambda_h * a_t^4 * \hat{\mu}_h^2 + 12 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 + 12 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 * \hat{\mu}_h^2 + 12 \\
& * \lambda_h^2 * a_t^2 * \hat{\mu}_h^2 - 12 * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - 12 * \frac{\kappa * a_t^3 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} \\
& + 12 * \frac{\kappa^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - 16 * \lambda_h * a_t^4 + 16 * \frac{\lambda_h * a_t^3 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} - 16 \\
& * \frac{\lambda_h * \kappa * a_t^2}{\varepsilon} + 18 * \frac{a_t^6}{\varepsilon} + 20 * \lambda_h * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + 21 \\
& * \frac{\kappa * a_t^4}{\varepsilon} - 24 * \lambda_h * \kappa * a_t^2 * \hat{\mu}_h^2 - 24 * \frac{\lambda_h * a_t^4}{\varepsilon} - 24 * \frac{\lambda_h * a_t^4 * \hat{\mu}_h^2}{\varepsilon} \\
& + 24 * \frac{\lambda_h^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - 27 * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 * \hat{\mu}_h^2 + 37 * \kappa * a_t^4 * \hat{\mu}_h^2 \\
& - 40 * \frac{\lambda_h * \kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + 54 * \frac{\kappa * a_t^4 * \hat{\mu}_h^2}{\varepsilon} - \frac{\lambda_\varphi * a_t^4 * \hat{\mu}_h^2}{\varepsilon} \Big) \\
& * M_s^{-2} * \varepsilon_{il} \varepsilon_{kn} \varepsilon_{jm} \cdot (H)_i (H)_j (H)_k (H^\dagger)_l (H^\dagger)_m (H^\dagger)_n
\end{aligned} \tag{30}$$

term30

$$\begin{aligned}
& - \left(\lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} - \lambda_h * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - 2 * \lambda_h * a_t^3 * \hat{\mu} + 2 * \lambda_h * \kappa * a_t^2 - 2 \right. \\
& \quad * \lambda_h^2 * a_t^2 + 2 * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - 2 * \frac{\lambda_h * a_t^3 * \hat{\mu}}{\varepsilon} + 2 * \frac{\lambda_h * \kappa * a_t^2}{\varepsilon} + 4 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} \\
& \quad * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 - 4 * \lambda_h * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - 4 * \frac{\lambda_h^2 * a_t^2}{\varepsilon} - 6 * \lambda_h * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 + 6 \\
& \quad * \lambda_h * a_t^4 * \hat{\mu}_h^2 - 6 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 - 6 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 * \hat{\mu}_h^2 + 6 * \lambda_h * \kappa * a_t^2 * \hat{\mu}_h^2 \\
& \quad - 6 * \lambda_h^2 * a_t^2 * \hat{\mu}_h^2 + 6 * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + 8 * \lambda_h * a_t^4 - 8 * \frac{\lambda_h * a_t^3 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} \\
& \quad \left. + 8 * \frac{\lambda_h * \kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + 12 * \frac{\lambda_h * a_t^4}{\varepsilon} + 12 * \frac{\lambda_h * a_t^4 * \hat{\mu}_h^2}{\varepsilon} - 12 * \frac{\lambda_h^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \right) \\
& \quad * M_s^{-2} * \varepsilon_{kn} \varepsilon_{jl} \varepsilon_{im} \cdot (H)_i (H)_j (H)_k (H^\dagger)_l (H^\dagger)_m (H^\dagger)_n
\end{aligned} \tag{31}$$

term31

$$\begin{aligned}
& + \left(2 * \lambda_h^2 * a_t^2 - 2 * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + 4 * \frac{\lambda_h^2 * a_t^2}{\varepsilon} + 6 * \lambda_h^2 * a_t^2 * \hat{\mu}_h^2 - 6 * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} \right. \\
& \quad \left. * a_t^2 * \hat{\mu}_h^2 + 12 * \frac{\lambda_h^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \right) * M_s^{-2} * \varepsilon_{jn} \varepsilon_{km} \varepsilon_{il} \cdot (H)_i (H)_j (H)_k (H^\dagger)_l (H^\dagger)_m (H^\dagger)_n
\end{aligned} \tag{32}$$

term32

$$\begin{aligned}
& + \left(\frac{1}{2} * \lambda_h^2 * a_t^2 - \frac{1}{2} * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{3}{2} * \lambda_h^2 * a_t^2 * \hat{\mu}_h^2 - \frac{3}{2} * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + 3 \right. \\
& \quad \left. * \frac{\lambda_h^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + \frac{\lambda_h^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} * \varepsilon_{jl} \varepsilon_{in} \varepsilon_{km} \cdot (H)_i (H)_j (H)_k (H^\dagger)_l (H^\dagger)_m (H^\dagger)_n
\end{aligned} \tag{33}$$

term33

$$\begin{aligned}
& + \left(\frac{1}{2} * \lambda_h^2 * a_t^2 - \frac{1}{2} * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{3}{2} * \lambda_h^2 * a_t^2 * \hat{\mu}_h^2 - \frac{3}{2} * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + 3 \right. \\
& \quad \left. * \frac{\lambda_h^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + \frac{\lambda_h^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} * \varepsilon_{jn} \varepsilon_{im} \varepsilon_{kl} \cdot (H)_i (H)_j (H)_k (H^\dagger)_l (H^\dagger)_m (H^\dagger)_n
\end{aligned} \tag{34}$$

term34

$$\begin{aligned}
& - \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) \\
& * M_s^{-2} * \varepsilon_{lm} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \delta_{ik} \delta_{nj} \cdot (H)_i (H^\dagger)_j (W_R)_{kl}^{\dot{\alpha}_1 \dot{\alpha}_2} (W_R)_{mn}^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& - \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) \\
& * M_s^{-2} * \varepsilon_{lm} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \delta_{ik} \delta_{nj} \cdot (H)_i (H^\dagger)_j (W_R)_{kl}^{\dot{\alpha}_1 \dot{\alpha}_2} (W_R)_{mn}^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& - \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * \varepsilon_{lm} \varepsilon^{\alpha_1 \alpha_2} \varepsilon^{\alpha_3 \alpha_4} \delta_{ik} \delta_{nj} \cdot (W_L)_{\alpha_1 \alpha_2, kl} (W_L)_{\alpha_3 \alpha_4, mn} (H)_i (H^\dagger)_j \\
& - \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * \varepsilon_{lm} \varepsilon^{\alpha_1 \alpha_4} \varepsilon^{\alpha_3 \alpha_2} \delta_{ik} \delta_{nj} \cdot (W_L)_{\alpha_1 \alpha_2, kl} (W_L)_{\alpha_3 \alpha_4, mn} (H)_i (H^\dagger)_j
\end{aligned} \tag{35}$$

term35

$$\begin{aligned}
& - \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) \\
& * M_s^{-2} * \varepsilon_{nk} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \delta_{im} \delta_{lj} \cdot (H)_i (H^\dagger)_j (W_R)_{kl}^{\dot{\alpha}_1 \dot{\alpha}_2} (W_R)_{mn}^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& - \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) \\
& * M_s^{-2} * \varepsilon_{nk} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \delta_{im} \delta_{lj} \cdot (H)_i (H^\dagger)_j (W_R)_{kl}^{\dot{\alpha}_1 \dot{\alpha}_2} (W_R)_{mn}^{\dot{\alpha}_3 \dot{\alpha}_4} \\
& - \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * \varepsilon_{nk} \varepsilon^{\alpha_1 \alpha_2} \varepsilon^{\alpha_3 \alpha_4} \delta_{im} \delta_{lj} \cdot (W_L)_{\alpha_1 \alpha_2, kl} (W_L)_{\alpha_3 \alpha_4, mn} (H)_i (H^\dagger)_j \\
& - \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * \varepsilon_{nk} \varepsilon^{\alpha_1 \alpha_4} \varepsilon^{\alpha_3 \alpha_2} \delta_{im} \delta_{lj} \cdot (W_L)_{\alpha_1 \alpha_2, kl} (W_L)_{\alpha_3 \alpha_4, mn} (H)_i (H^\dagger)_j
\end{aligned} \tag{36}$$

term36

$$0 \tag{37}$$

term37

$$0 \tag{38}$$

term38

$$\begin{aligned}
& - \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} - \frac{5}{8} * a_t^2 \right) * M_s^{-2} \\
& * (y_e)^{vw} (y_e^\dagger)^{wu} \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (L)_{\alpha_1, i}^u (H)_{\alpha_2, j}^{\dot{\alpha}_1} (H^\dagger)_k (L^\dagger)_l^{\dot{\alpha}_2, v}
\end{aligned} \tag{39}$$

term39

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& * (y_d)^{vw} (y_d^\dagger)^{wu} \varepsilon_{jl} \varepsilon_{ik} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (Q)_{\alpha_1, i, a}^u (H)_{\alpha_2, j}^{\dot{\alpha}_1} (H^\dagger)_k (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, v}
\end{aligned} \tag{40}$$

term40

$$\begin{aligned}
& - \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} - \frac{5}{8} * a_t^2 \right) * M_s^{-2} * (y_e)^{uv} (y_e^\dagger)^{uw} \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \\
& \cdot (e_{\mathbb{C}})_\alpha^u (H)_{\alpha_2, i}^{\dot{\alpha}_1} (H^\dagger)_j (e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, v}
\end{aligned} \tag{41}$$

term41

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& * (y_d)^{uv} (y_d^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (d_{\mathbb{C}})_\alpha^u (H)_{\alpha_2, i}^{\dot{\alpha}_1} (H^\dagger)_j (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, v}
\end{aligned} \tag{42}$$

term42

$$\begin{aligned}
& + \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& * (y_u)^{uv} (y_u^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (u_{\mathbb{C}})_\alpha^u (H)_{\alpha_2, i}^{\dot{\alpha}_1} (H^\dagger)_j (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, v}
\end{aligned} \tag{43}$$

term43

$$\begin{aligned}
& - \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} - \frac{5}{8} * a_t^2 \right) * M_s^{-2} \\
& * (y_e)^{vw} (y_e^\dagger)^{wu} \varepsilon_{jl} \varepsilon_{ik} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (L)_{\alpha_1, i}^u (H)_j (H^\dagger)_{\alpha_2, k}^{\dot{\alpha}_1} (L^\dagger)_l^{\dot{\alpha}_2, v}
\end{aligned} \tag{44}$$

term44

$$\begin{aligned}
& + \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& * (y_d)^{vw} (y_d^\dagger)^{wu} \varepsilon_{jl} \varepsilon_{ik} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (Q)_{\alpha_1, i, a}^u (H)_j (H^\dagger)_{\alpha_2, k}^{\dot{\alpha}_1} (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, v}
\end{aligned} \tag{45}$$

term45

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& * (y_u)^{uv} (y_u^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (u_{\mathbb{C}})_\alpha^u (H)_i (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_1} (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, v}
\end{aligned} \tag{46}$$

term46

$$\begin{aligned}
& - \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} - \frac{5}{8} * a_t^2 \right) * M_s^{-2} * (y_e)^{uv} (y_e^\dagger)^{uw} \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \\
& \cdot (e_{\mathbb{C}})_\alpha^u (H)_i (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_1} (e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, v}
\end{aligned} \tag{47}$$

term47

$$\begin{aligned}
& + \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& * (y_d)^{uv} (y_d^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (d_{\mathbb{C}})_\alpha^u (H)_i (H^\dagger)_{\alpha_2, j}^{\dot{\alpha}_1} (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, v}
\end{aligned} \tag{48}$$

term48

$$+ \left(\frac{1}{4} * a_t^2 \right) * M_s^{-2} * (y_e)^{vw} (y_e^\dagger)^{wu} \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (L)_{\alpha_1 \alpha_2, i}^{\dot{\alpha}_1, u} (H)_j (H^\dagger)_k (L^\dagger)_l^{\dot{\alpha}_2, v} \quad (49)$$

term49

$$- \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} * (y_d)^{vw} (y_d^\dagger)^{wu} \varepsilon_{ik} \varepsilon_{jl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (Q)_{\alpha_1 \alpha_2, i, a}^{\dot{\alpha}_1, u} (H)_j (H^\dagger)_k (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, v} \quad (50)$$

term50

$$- \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} * (y_u)^{vw} (y_u^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (u_{\mathbb{C}})_{\alpha_1, ab}^u (H)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)_{\alpha_2, c}^{\dot{\alpha}_1 \dot{\alpha}_2, v} \quad (51)$$

term51

$$+ \left(\frac{1}{4} * a_t^2 \right) * M_s^{-2} * (y_e)^{vw} (y_e^\dagger)^{uw} \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (e_{\mathbb{C}})_{\alpha_1}^u (H)_i (H^\dagger)_j (e_{\mathbb{C}}^\dagger)_{\alpha_2}^{\dot{\alpha}_1 \dot{\alpha}_2, v} \quad (52)$$

term52

$$+ \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} * (y_d)^{vw} (y_d^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (d_{\mathbb{C}})_{\alpha_1, ab}^u (H)_i (H^\dagger)_j (d_{\mathbb{C}}^\dagger)_{\alpha_2, c}^{\dot{\alpha}_1 \dot{\alpha}_2, v} \quad (53)$$

term53

$$+ \left(\frac{1}{4} * a_t^2 \right) * M_s^{-2} * (y_e)^{vw} (y_e^\dagger)^{wu} \varepsilon_{jl} \varepsilon_{ik} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (L)_{\alpha_1, i}^u (H)_j (H^\dagger)_k (L^\dagger)_{\alpha_2, l}^{\dot{\alpha}_1 \dot{\alpha}_2, v} \quad (54)$$

term54

$$- \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} * (y_d)^{vw} (y_d^\dagger)^{wu} \varepsilon_{ik} \varepsilon_{jl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (Q)_{\alpha_1, i, a}^u (H)_j (H^\dagger)_k (Q^\dagger)_{\alpha_2, l, bc}^{\dot{\alpha}_1 \dot{\alpha}_2, v} \quad (55)$$

term55

$$+ \left(\frac{1}{4} * a_t^2 \right) * M_s^{-2} * (y_e)^{vw} (y_e^\dagger)^{uw} \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (e_{\mathbb{C}})_{\alpha_1 \alpha_2}^{\dot{\alpha}_1, u} (H)_i (H^\dagger)_j (e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, v} \quad (56)$$

term56

$$+ \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} * (y_d)^{vw} (y_d^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (d_{\mathbb{C}})_{\alpha_1 \alpha_2, ab}^{\dot{\alpha}_1, u} (H)_i (H^\dagger)_j (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, v} \quad (57)$$

term57

$$+ \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} * (y_u)^{vw} (y_u^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (u_{\mathbb{C}})^{\dot{\alpha}_1, u}_{\alpha_1 \alpha_2, ab} (H)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, v}_c \quad (58)$$

term58

$$- \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{8} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\ * (y_e)^{vw} (y_e^\dagger)^{wu} \varepsilon_{jl} \varepsilon_{ik} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (L)^u_{\alpha_1, i} (H)_j (H^\dagger)_k (L^\dagger)^{\dot{\alpha}_1 \dot{\alpha}_2, v}_{\alpha_2, l} \quad (59)$$

term59

$$- \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\ * (y_d)^{vw} (y_d^\dagger)^{wu} \varepsilon_{jl} \varepsilon_{ik} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (Q)^u_{\alpha_1, i, a} (H)_j (H^\dagger)_k (Q^\dagger)^{\dot{\alpha}_1 \dot{\alpha}_2, v}_{\alpha_2, l, bc} \quad (60)$$

term60

$$+ \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\ * (y_u)^{vw} (y_u^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (u_{\mathbb{C}})^{\dot{\alpha}_1, u}_{\alpha_1 \alpha_2, ab} (H)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, v}_c \quad (61)$$

term61

$$- \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{8} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\ * (y_e)^{vw} (y_e^\dagger)^{uw} \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (e_{\mathbb{C}})^{\dot{\alpha}_1, u}_{\alpha_1 \alpha_2} (H)_i (H^\dagger)_j (e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, v} \quad (62)$$

term62

$$+ \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\ * (y_d)^{vw} (y_d^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (d_{\mathbb{C}})^{\dot{\alpha}_1, u}_{\alpha_1 \alpha_2, ab} (H)_i (H^\dagger)_j (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, v}_c \quad (63)$$

term63

$$- \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{8} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\ * (y_e)^{vw} (y_e^\dagger)^{wu} \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (L)^{\dot{\alpha}_1, u}_{\alpha_1 \alpha_2, i} (H)_j (H^\dagger)_k (L^\dagger)^{\dot{\alpha}_2, v}_l \quad (64)$$

term64

$$+ \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\ * (y_d)^{vw} (y_d^\dagger)^{wu} \varepsilon_{ik} \varepsilon_{jl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (Q)^{\dot{\alpha}_1, u}_{\alpha_1 \alpha_2, i, a} (H)_j (H^\dagger)_k (Q^\dagger)^{\dot{\alpha}_2, v}_{l, bc} \quad (65)$$

term65

$$\begin{aligned}
& - \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{8} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_e)^{wv} (y_e^\dagger)^{uw} \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (e_{\mathbb{C}})_{\alpha_1}^u (H)_i (H^\dagger)_j (e_{\mathbb{C}}^\dagger)_{\alpha_2}^{\dot{\alpha}_1 \dot{\alpha}_2, v}
\end{aligned} \tag{66}$$

term66

$$\begin{aligned}
& + \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_d)^{wv} (y_d^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (d_{\mathbb{C}})_{\alpha_1, ab}^u (H)_i (H^\dagger)_j (d_{\mathbb{C}}^\dagger)_{\alpha_2, c}^{\dot{\alpha}_1 \dot{\alpha}_2, v}
\end{aligned} \tag{67}$$

term67

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_u)^{wv} (y_u^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (u_{\mathbb{C}})_{\alpha_1, ab}^u (H)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)_{\alpha_2, c}^{\dot{\alpha}_1 \dot{\alpha}_2, v}
\end{aligned} \tag{68}$$

term68

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& * (y_d)^{wv} (y_u^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (u_{\mathbb{C}})_{\alpha_1, ab}^u (H)_{\alpha_2, i}^{\dot{\alpha}_1} (H)_j (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, v}
\end{aligned} \tag{69}$$

term69

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& * (y_d)^{wv} (y_u^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (u_{\mathbb{C}})_{\alpha_1, ab}^u (H)_{\alpha_2, i}^{\dot{\alpha}_1} (H)_j (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, v}
\end{aligned} \tag{70}$$

term70

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& * (y_u)^{wv} (y_d^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (d_{\mathbb{C}})_{\alpha_1, ab}^u (H^\dagger)_{\alpha_2, i}^{\dot{\alpha}_1} (H^\dagger)_j (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, v}
\end{aligned} \tag{71}$$

term71

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& * (y_u)^{wv} (y_d^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (d_{\mathbb{C}})_{\alpha_1, ab}^u (H^\dagger)_{\alpha_2, i}^{\dot{\alpha}_1} (H^\dagger)_j (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, v}
\end{aligned} \tag{72}$$

term72

$$+ \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} * (y_u)^{wv} (y_d^\dagger)^{uw} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (d_{\mathbb{C}})_{\alpha_1, ab}^u (H^\dagger)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)_{\alpha_2, c}^{\dot{\alpha}_1 \dot{\alpha}_2, v} \tag{73}$$

term73

$$+ \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} * (y_d)^{wv} (y_u^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (u_{\mathbb{C}})^u_{\alpha_1, ab} (H)_i (H)_j (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_1 \dot{\alpha}_2, v}_{\alpha_2, c} \quad (74)$$

term74

$$- \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} * (y_d)^{wv} (y_u^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (u_{\mathbb{C}})^{\dot{\alpha}_1, u}_{\alpha_1 \alpha_2, ab} (H)_i (H)_j (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, v}_c \quad (75)$$

term75

$$- \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} * (y_u)^{wv} (y_d^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (d_{\mathbb{C}})^{\dot{\alpha}_1, u}_{\alpha_1 \alpha_2, ab} (H^\dagger)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, v}_c \quad (76)$$

term76

$$\begin{aligned} & - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\ & * (y_u)^{wv} (y_d^\dagger)^{uw} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (d_{\mathbb{C}})^{\dot{\alpha}_1, u}_{\alpha_1 \alpha_2, ab} (H^\dagger)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, v}_c \end{aligned} \quad (77)$$

term77

$$\begin{aligned} & - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\ & * (y_d)^{wv} (y_u^\dagger)^{uw} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (u_{\mathbb{C}})^{\dot{\alpha}_1, u}_{\alpha_1 \alpha_2, ab} (H)_i (H)_j (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, v}_c \end{aligned} \quad (78)$$

term78

$$\begin{aligned} & + \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\ & * (y_d)^{wv} (y_u^\dagger)^{uw} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (u_{\mathbb{C}})^u_{\alpha_1, ab} (H)_i (H)_j (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_1 \dot{\alpha}_2, v}_{\alpha_2, c} \end{aligned} \quad (79)$$

term79

$$\begin{aligned} & - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\ & * (y_u)^{wv} (y_d^\dagger)^{uw} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (d_{\mathbb{C}})^u_{\alpha_1, ab} (H^\dagger)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_1 \dot{\alpha}_2, v}_{\alpha_2, c} \end{aligned} \quad (80)$$

term80

$$\begin{aligned} & - \left(\frac{1}{8} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * d * a_t^2 - \frac{1}{4} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\ & * (y_e)^{wx} (y_e^\dagger)^{xv} (y_e^\dagger)^{uw} \varepsilon_{il} \varepsilon_{jk} \varepsilon^{\alpha_1 \alpha_2} \cdot (e_{\mathbb{C}})^u_{\alpha_1} (L)^v_{\alpha_2, i} (H)_j (H^\dagger)_k (H^\dagger)_l \end{aligned} \quad (81)$$

term81

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_d)^{wx} (y_d^\dagger)^{xv} (y_d^\dagger)^{uw} \varepsilon_{il} \varepsilon_{jk} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \cdot (d_{\mathbb{C}})_{\alpha_1, ab}^u (Q)_{\alpha_2, i, c}^v (H)_j (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{82}$$

term82

$$\begin{aligned}
& - \left(\frac{1}{8} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * d * a_t^2 - \frac{1}{4} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_e)^{vw} (y_e)^{xu} (y_e^\dagger)^{wx} \varepsilon_{il} \varepsilon_{jk} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_i (H)_j (H^\dagger)_k (e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_1, u} (L^\dagger)_l^{\dot{\alpha}_2, v}
\end{aligned} \tag{83}$$

term83

$$\begin{aligned}
& + \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_d)^{wu} (y_d)^{vx} (y_d^\dagger)^{xw} \varepsilon_{il} \varepsilon_{jk} \varepsilon_{abc} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_i (H)_j (H^\dagger)_k (d_{\mathbb{C}}^\dagger)_a^{\dot{\alpha}_1, u} (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, v}
\end{aligned} \tag{84}$$

term84

$$\begin{aligned}
& + \left(\frac{1}{8} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * d * a_t^2 - \frac{1}{4} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_e)^{wu} (y_e)^{vx} (y_e^\dagger)^{xw} \varepsilon_{ik} \varepsilon_{jl} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_i (H)_j (H^\dagger)_k (e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_1, u} (L^\dagger)_l^{\dot{\alpha}_2, v}
\end{aligned} \tag{85}$$

term85

$$\begin{aligned}
& + \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_d)^{vw} (y_d)^{xu} (y_d^\dagger)^{wx} \varepsilon_{jl} \varepsilon_{ik} \varepsilon_{abc} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_i (H)_j (H^\dagger)_k (d_{\mathbb{C}}^\dagger)_a^{\dot{\alpha}_1, u} (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, v}
\end{aligned} \tag{86}$$

term86

$$\begin{aligned}
& + \left(\frac{1}{8} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * d * a_t^2 - \frac{1}{4} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_e)^{wx} (y_e^\dagger)^{uw} (y_e^\dagger)^{xv} \varepsilon_{jl} \varepsilon_{ik} \varepsilon^{\alpha_1 \alpha_2} \cdot (e_{\mathbb{C}})_{\alpha_1}^u (L)_{\alpha_2, i}^v (H)_j (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{87}$$

term87

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_d)^{wx} (y_d^\dagger)^{xv} (y_d^\dagger)^{uw} \varepsilon_{jl} \varepsilon_{ik} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \cdot (d_{\mathbb{C}})_{\alpha_1, ab}^u (Q)_{\alpha_2, i, c}^v (H)_j (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{88}$$

term88

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& * (y_u)^{vw} (y_u^\dagger)^{wu} \varepsilon_{ij} \varepsilon_{lk} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (Q)_{\alpha_1, i, a}^u (H)_{\alpha_2, j}^{\dot{\alpha}_1} (H^\dagger)_k (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, v}
\end{aligned} \tag{89}$$

term89

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& * (y_u)^{vw} (y_u^\dagger)^{wu} \varepsilon_{ij} \varepsilon_{kl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (Q)_{\alpha_1, i, a}^u (H)_j (H^\dagger)_{\alpha_2, k}^{\dot{\alpha}_1} (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, v}
\end{aligned} \tag{90}$$

term90

$$+ \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} * (y_u)^{vw} (y_u^\dagger)^{wu} \varepsilon_{ij} \varepsilon_{lk} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (Q)_{\alpha_1 \alpha_2, i, a}^{\dot{\alpha}_1, u} (H)_j (H^\dagger)_k (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, v} \tag{91}$$

term91

$$- \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} * (y_u)^{vw} (y_u^\dagger)^{wu} \varepsilon_{ij} \varepsilon_{kl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (Q)_{\alpha_1, i, a}^u (H)_j (H^\dagger)_k (Q^\dagger)_{\alpha_2, l, bc}^{\dot{\alpha}_1 \dot{\alpha}_2, v} \tag{92}$$

term92

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_u)^{vw} (y_u^\dagger)^{wu} \varepsilon_{ij} \varepsilon_{kl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (Q)_{\alpha_1, i, a}^u (H)_j (H^\dagger)_k (Q^\dagger)_{\alpha_2, l, bc}^{\dot{\alpha}_1 \dot{\alpha}_2, v}
\end{aligned} \tag{93}$$

term93

$$\begin{aligned}
& + \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_u)^{vw} (y_u^\dagger)^{wu} \varepsilon_{ji} \varepsilon_{kl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (Q)_{\alpha_1 \alpha_2, i, a}^{\dot{\alpha}_1, u} (H)_j (H^\dagger)_k (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, v}
\end{aligned} \tag{94}$$

term94

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_u)^{uw} (y_u)^{xv} (y_u^\dagger)^{wx} \varepsilon_{ij} \varepsilon_{kl} \varepsilon_{cab} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_i (H^\dagger)_j (H^\dagger)_k (Q^\dagger)_{l, ab}^{\dot{\alpha}_1, u} (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, v}
\end{aligned} \tag{95}$$

term95

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_u)^{wv} (y_d)^{ux} (y_d^\dagger)^{xw} \varepsilon_{il} \varepsilon_{kj} \varepsilon_{cab} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_i (H^\dagger)_j (H^\dagger)_k (Q^\dagger)_{l, ab}^{\dot{\alpha}_1, u} (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, v}
\end{aligned} \tag{96}$$

term96

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_d)^{wx} (y_u^\dagger)^{vw} (y_d^\dagger)^{xu} \varepsilon_{il} \varepsilon_{jk} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \cdot (Q)_{\alpha_1, i, a}^u (u_{\mathbb{C}})_{\alpha_2, bc}^v (H)_j (H)_k (H^\dagger)_l
\end{aligned} \tag{97}$$

term97

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_u)^{wx} (y_u^\dagger)^{vw} (y_u^\dagger)^{xu} \varepsilon_{ki} \varepsilon_{jl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \cdot (Q)_{\alpha_1, i, a}^u (u_{\mathbb{C}})_{\alpha_2, bc}^v (H)_j (H)_k (H^\dagger)_l
\end{aligned} \tag{98}$$

term98

$$\begin{aligned}
& + \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_u)^{wx} (y_u^\dagger)^{vw} (y_u^\dagger)^{xu} \varepsilon_{ij} \varepsilon_{kl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \cdot (Q)_{\alpha_1, i, a}^u (u_{\mathbb{C}})_{\alpha_2, bc}^v (H)_j (H)_k (H^\dagger)_l
\end{aligned} \tag{99}$$

term99

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_u)^{uw} (y_u)^{xv} (y_u^\dagger)^{wx} \varepsilon_{ik} \varepsilon_{jl} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \cdot (H)_i (H^\dagger)_j (H^\dagger)_k (Q^\dagger)_{l, ab}^{\dot{\alpha}_1, u} (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, v}
\end{aligned} \tag{100}$$

term100

$$\begin{aligned}
& + \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_d)^{wx} (y_u^\dagger)^{vw} (y_d^\dagger)^{xu} \varepsilon_{jk} \varepsilon_{il} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \cdot (Q)_{\alpha_1, i, a}^u (u_{\mathbb{C}})_{\alpha_2, bc}^v (H)_j (H)_k (H^\dagger)_l
\end{aligned} \tag{101}$$

term101

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_u)^{wv} (y_d)^{ux} (y_d^\dagger)^{xw} \varepsilon_{il} \varepsilon_{jk} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \cdot (H)_i (H^\dagger)_j (H^\dagger)_k (Q^\dagger)_{l, ab}^{\dot{\alpha}_1, u} (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, v}
\end{aligned} \tag{102}$$

term102

$$\begin{aligned}
& + \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_u)^{wx} (y_u^\dagger)^{xv} (y_d^\dagger)^{uw} \varepsilon_{ij} \varepsilon_{lk} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \cdot (d_{\mathbb{C}})_{\alpha_1, ab}^u (Q)_{\alpha_2, i, c}^v (H)_j (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{103}$$

term103

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_u)^{vw} (y_d)^{xu} (y_u^\dagger)^{wx} \varepsilon_{ji} \varepsilon_{kl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \cdot (H)_i (H)_j (H^\dagger)_k (d_{\mathbb{C}}^\dagger)_a^{\dot{\alpha}_1, u} (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, v}
\end{aligned} \tag{104}$$

term104

$$\begin{aligned}
& + \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_u)^{vw} (y_d)^{xu} (y_u^\dagger)^{wx} \varepsilon_{ji} \varepsilon_{kl} \varepsilon_{abc} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \cdot (H)_i (H)_j (H^\dagger)_k (d_{\mathbb{C}}^\dagger)_a^{\dot{\alpha}_1, u} (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, v}
\end{aligned} \tag{105}$$

term105

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& * (y_u)^{wx} (y_u^\dagger)^{xv} (y_d^\dagger)^{uw} \varepsilon_{ji} \varepsilon_{kl} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \cdot (d_{\mathbb{C}})_\alpha^u (Q)_{\alpha_2, i, c}^v (H)_j (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{106}$$